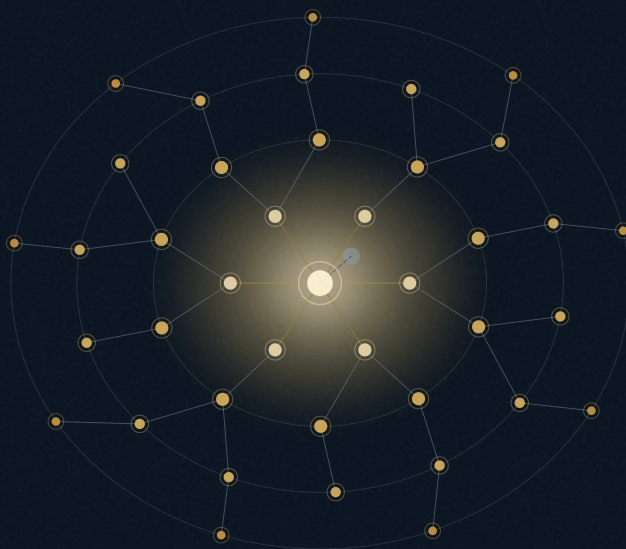




KRONOS FUSION ENERGY



PAPER 1.5 OF THE 2026 KRONOS SET

AI, ML & Quantum Control

Certifiable Real-Time Control

Physics-informed AI and quantum technologies for certifiable real-time control
of compact fusion generators — three tiers, held apart.

Layered twin · 0/5000 escapes clamped · quantum: no crossover this decade

Explore it live — [3-D model](#) · [physics validator](#) · [film series](#)

THE OFFICIAL RECORD

What this document is, and how to read its numbers

This is a combined editorial. It gathers, in one volume, the two design studies Kronos Fusion Energy released in 2026 — a compact spherical-tokamak tritium/helium-3 breeder and a deuterium–helium-3 tandem-mirror generator — together with the intellectual-property estate, the people, and, in the internal edition, the full commercial case. The scientific text is the same text that appears in the arXiv preprints and the journal submissions; the editorial adds the apparatus a reader needs to hold the whole program at once, and nothing in the physics is altered to fit it.

THE HONESTY STANCE IS THE ASSET

Every headline quantity carries an **evidence class** — DERIVED RECOMPUTED MEASURED REQUIREMENT — and, where a number is a modelled extrapolation rather than a demonstrated value, it is labelled as such and its downside is shown next to it. Several quantities in the underlying corpus moved after first publication; where a value has been withdrawn or restated, the record says so plainly. This is deliberate: a diligence team will find the load-bearing assumptions, so the document surfaces them first.

TITLE	The Kronos Fleet — Hyperion · Aegis · MetroVolt: A Combined Design & Commercialization Study
AUTHOR	Priyanca Ford, Founder & CEO, on behalf of Kronos Fusion Energy
EDITION	2026 Combined Editorial · first issue
COMPANION WORKS	The five-paper arXiv drop — (1) Breeder, (2) Burner, (3) REBCO magnet & tape, (4) Direct Energy Conversion, (5) AI/ML/Quantum control — with matching numbered Zenodo deposits, plus the IOP journal and IAEA-FEC submissions. Patents were filed before the drop.
NAMING	Hyperion = the breeder; Aegis (defense) and MetroVolt (data-center) = one generator in two housings. Internal mode letters (D1, L) do not appear on public surfaces.
STATUS	Conceptual design & simulation study. Nothing herein is a construction commitment.



HOW TO READ THIS VOLUME

The fleet at a glance, and the way its numbers are marked

This volume opens with *Orientation*, presents the five research papers as a bound sequence, and closes with the *Apparatus*. *Foundations* sets out the three general results both machines rest on. *Paper One* is the Hyperion breeder; *Paper Two* the Aegis / MetroVolt generator; *Papers Three, Four and Five* are the enabling science — high-field REBCO magnets and tape, direct energy conversion, and the AI / ML / quantum control stack. A *Digital Twin & 3-D Model* part follows, then the *Environmental* profile. The *Economics* and levelized cost and the *Business Case* and diligence record appear in the internal edition only; the *Simulations* proof record and the Apparatus — patent portfolio, team, limitations, and sources — close both editions.

THE FLEET AT A GLANCE

	HYPERION	AEGIS	METROVOLT
Role	Strategic-isotope foundry	Defense installation power	Campus power
Machine	Spherical tokamak	Tandem mirror	Tandem mirror
Fuel	Deuterium–tritium	Deuterium–helium-3	Deuterium–helium-3
Delivers	Tritium · ³ He · 14 MeV n	Resilient installation power	Firm campus power
Physics bar	Fusion gain only	Closure (gates named)	Closure + lunar ³ He
In the fleet	Breeds the fuel	Proves the generator	Commercial destination

HOW THE NUMBERS ARE MARKED

Every headline quantity carries an evidence class — **DERIVED** from first principles, **RECOMPUTED** from raw data, **MEASURED** in hardware, or a **REQUIREMENT** yet to be met. Financial figures carry a case label and are confined to the internal (confidential) edition. Values that moved after first publication are shown as withdrawn or restated rather than quietly changed. A capability you can evaluate is one whose limits are on the page.

ABSTRACT

AI, ML & Quantum Control — Certifiable Real-Time Control

Real-time control of a nuclear-regulated fusion generator is treated as a layered digital-twin architecture in which physics-informed machine learning is deployable now — but only inside a deterministically clamped safe set. A control-barrier filter permits 0 safety-set escapes across 5000 trajectories against 5000 unfiltered, backed by an analytic proof. Quantum sensing is a near-term diagnostic tier; quantum *computation* is an offline calibration tier with no resource crossover this decade. Every miss is reported, including the equilibrium network's 1.4% against a 1% target.

Companion deposit (code & data, reproducible under CC BY 4.0): DOI [10.5281/zenodo.22132484](https://doi.org/10.5281/zenodo.22132484) · Zenodo community *kronos_fusion_energy*.

Explore it live: [interactive 3-D model](#) · [run the physics validator](#) · [companion film series](#) (the deposited solvers, in the browser).

Keywords: [digital twin](#) [physics-informed ML](#) [certifiable control](#) [quantum sensing](#) [safe-set clamp](#)

CONTENTS

Table of Contents

What this document is, and how to read its numbers	1
The fleet at a glance, and the way its numbers are marked	2
AI, ML & Quantum Control — Certifiable Real-Time Control	3
Symbols, units, and the names of things	5
§ 1 Three tiers, held apart	6
§ 2 Tier 1 — physics-informed AI, deployable now if clamped	7
§ 3 Tier 3 — quantum computation, an offline calibration tier	8
§ 4 Reporting the misses	9

PORTFOLIO, PEOPLE & RECORD

The patent portfolio	11
Founding partners, board, advisors & auditors	12
List of Figures	14
Sources	15
How this document was made	17
Notice, status, and distribution	18
One programme, in five papers	19

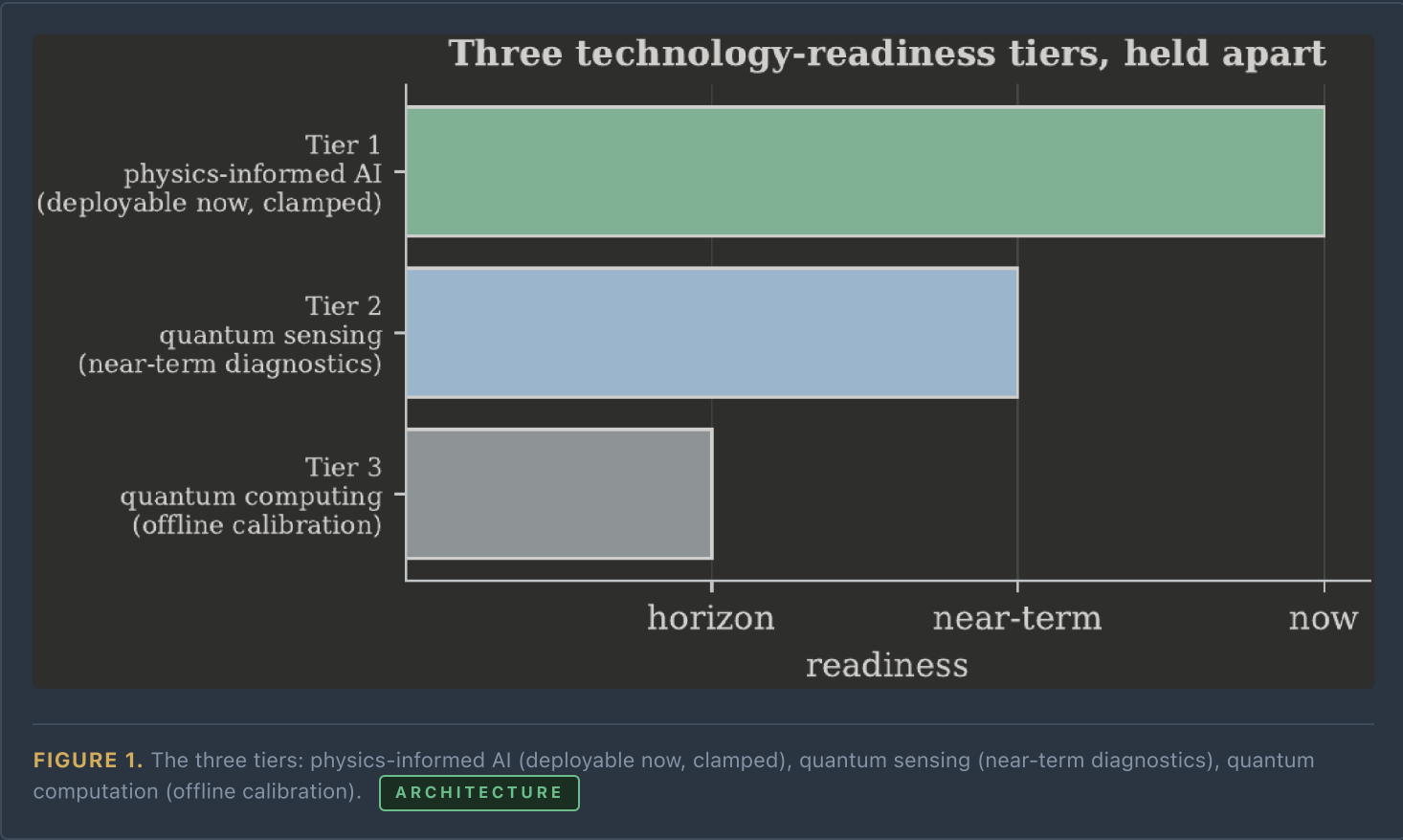
NOMENCLATURE

Symbols, units, and the names of things

Q	plasma (fusion) gain, $P_{\text{fus}}/P_{\text{aux}}$
Q_E	engineering gain, net electric out / recirculating in
H_{98}	confinement enhancement over the IPB98(y,2) scaling
Z_{eff}	effective ion charge
c_{ee}	electron–electron bremsstrahlung leading coefficient, 2.120022 (exact)
τ_E	energy confinement time
I_p	plasma current
R_0, a	major radius, minor radius
κ, δ	elongation, triangularity
β_N	normalized plasma beta (Troyon-normalized)
TBR	tritium breeding ratio
DEC	direct energy conversion
CF	plant capacity factor (availability)
$FOAK/NOAK/BOAK$	first / n th / best of a kind
<i>Hyperion</i>	the ST breeder (internal: Mode D2)
<i>Aegis</i>	the generator, defense/installation housing (internal: Mode M)
<i>MetroVolt</i>	the generator, data-center housing (Mode M)

Three tiers, held apart

A nuclear-regulated fusion generator must be controlled across ten orders of magnitude in time — microsecond magnet protection through month-scale fuel logistics — and every automated actuation must be bounded, auditable, and reproducible. Machine learning and quantum technologies are often invoked for this without saying what is deployable now versus what is a decade away. We hold three technology-readiness tiers strictly apart, and report each with its evidence *and* its misses.



Tier 1 — physics-informed AI, deployable now if clamped

The load-bearing requirement is that a learned controller cannot drive the machine out of its safe operating envelope. A robust control-barrier-function filter provides this with an analytic forward-invariance guarantee: over 5000 Monte-Carlo trajectories, 0 escape the safe set with the filter against 5000 of 5000 without it. Around it, a Bayesian-optimization surrogate reaches leave-one-out $R^2 = 0.987$. Physics-informed AI is deployable in the loop *because* its error is bounded and deterministically clamped.

WHY THIS IS THE LOAD-BEARING RESULT

A learned controller can pursue performance while a proven clamp guarantees the machine stays in the safe envelope — zero safe-set escapes with the filter, backed by proof, not by a training score.

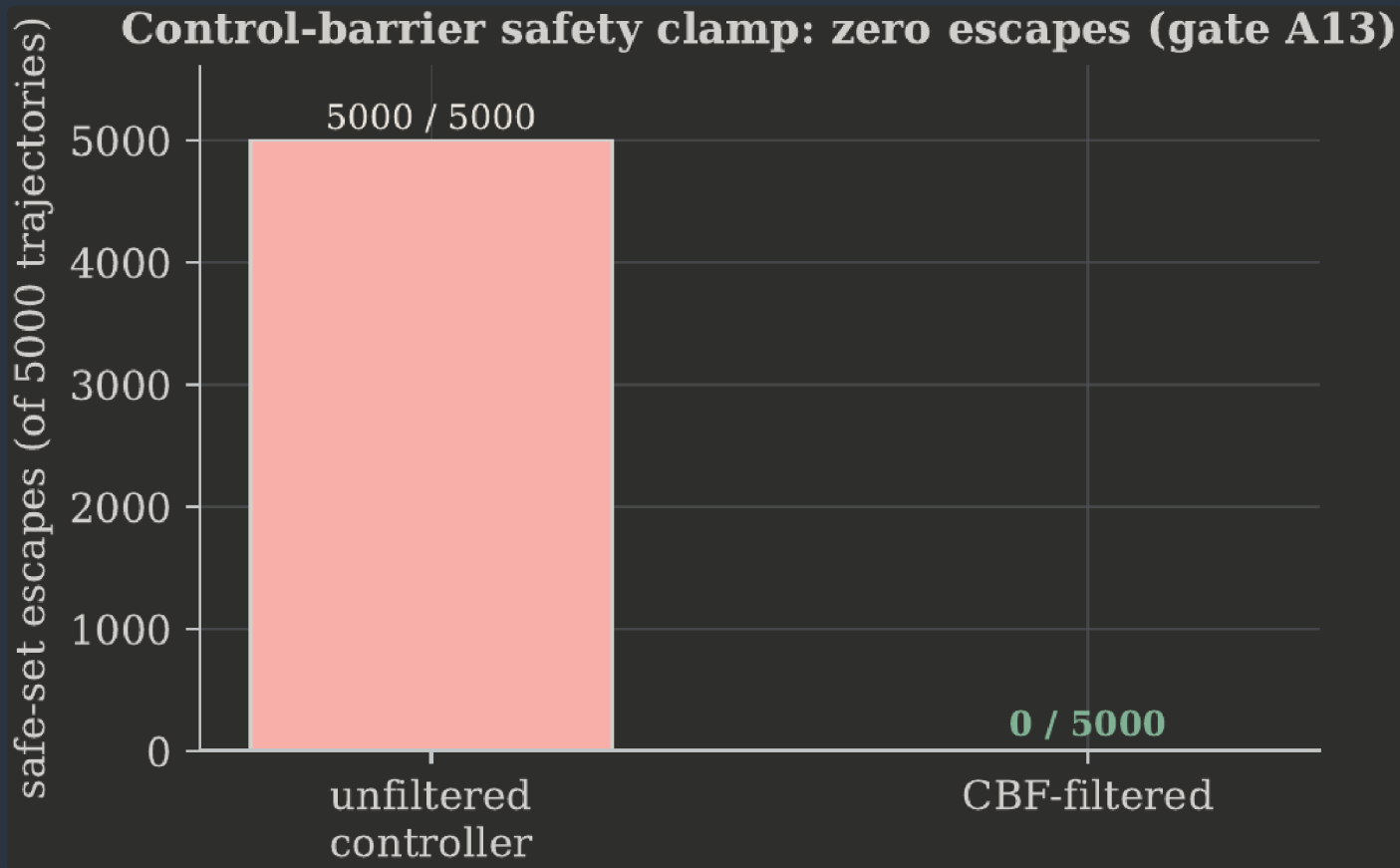
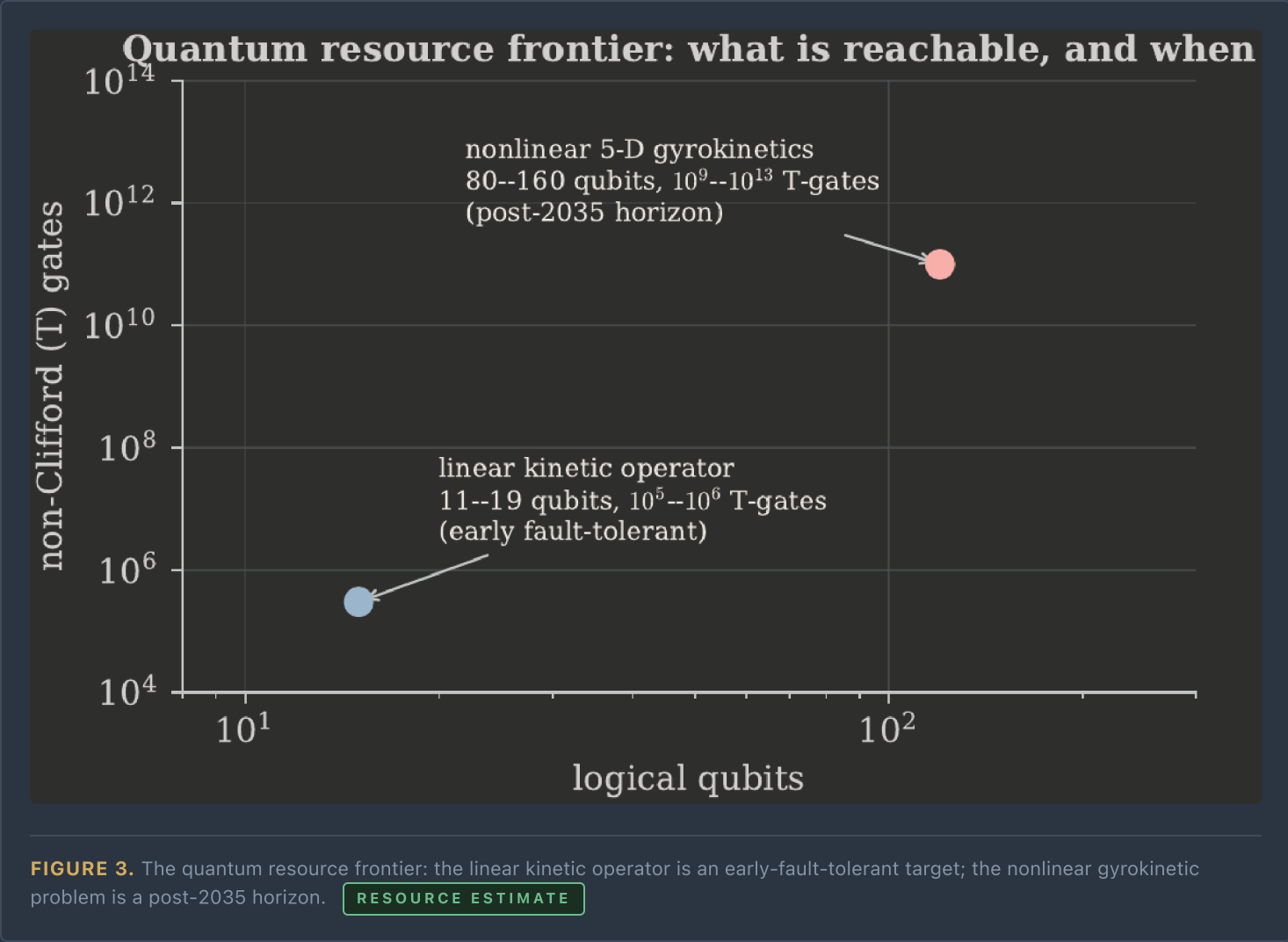


FIGURE 2. The control-barrier safety clamp: 0 of 5000 trajectories escape the safe set with the filter, against 5000 unfiltered. MONTE-CARLO

Tier 3 — quantum computation, an offline calibration tier

For fusion, quantum computation is an offline calibration tier with no resource crossover this decade. Landau damping reproduces on a 5-qubit circuit to a few percent, and variational quantum chemistry reaches chemical accuracy on the fusion-alloy fragments — but the honest resource split is load-bearing: the linear kinetic operator needs 11–19 logical qubits, and the nonlinear gyrokinetic problem that would most help the machine is fault-tolerant-horizon. Noisy hardware today shows no advantage; the classical baseline is cheaper at every demonstrated scale.



Reporting the misses

We report every missed pre-registered target rather than reframing it. The physics-informed equilibrium network was intended to become a fast native solver and did not: it reached **1.4%** relative- L^2 error against a 1% goal. Keeping this negative in the record is part of the method. The contribution is a certifiable, resource-honest system and method for compact-fusion control — not new plasma physics.

REPORTED, NOT REFRAMED

A P P A R A T U S

Portfolio, People & Record

*The intellectual-property estate, the people who built it, the limitations
that gate it, and the sources it rests on.*

1

GRANTED PATENT

4

2026 PROVISIONALS

40

TEAM MEMBERS

27

2022 PROVISIONALS

INTELLECTUAL PROPERTY

The patent portfolio

The estate spans a granted high-field magnet patent, pending utility and 2026 provisional filings that map to the two products, a digital-twin control provisional, a registered trademark application, and the original 2022 provisional family. Every patentable disclosure in the five-paper arXiv drop was protected **before** publication: the breeder and burner provisionals were filed 1–2 August, and a publication-gap omnibus filing the evening before the drop swept up the DEC, plasma-control, and REBCO-winding matter of the three otherwise-unprotected papers. Patent numbers and application serials are matters of public record; claim scope is summarised, not reproduced.

GRANTED & PENDING UTILITY

US 12,009,112	High-field tilted graded-REBCO magnet architecture (KRONO-002A) · app. 17/878,550	GRANTED
US 17/878,507	Core device / method (KRONO-001A) · utility	PENDING

2026 PROVISIONAL FILINGS — MAPPED TO THE PRODUCTS

64/124,209	Hyperion — spherical-tokamak tritium / helium-3 foundry · breeder · filed Aug 2026	FILED
64/124,220	Aegis / MetroVolt — synchrotron-cutoff, fuel-selection & plug-winding · generator · filed Aug 2026	FILED
64/115,167	KRONOS-CTRL digital-twin plant control · provisional · filed Jul 2026	FILED
64/005,440	Integrated spherical-tokamak fusion architecture · early integrated-architecture filing · filed Mar 2026	FILED
64/105,530	Negative-triangularity spherical-tokamak architecture · foundational ST filing · filed Jul 2026	FILED
64/128,097	Publication-gap omnibus — quasineutral two-species expander DEC · provenance-gated / latency-split plasma control · as-built high-field winding & conductor architecture · filed 7 Aug 2026, the evening before the arXiv drop	FILED

TRADEMARK & ORIGIN

FTK-98801894	Kronos Fusion Energy — trademark application	FILED
KR-2022-★	Original provisional family (KR-2022-00001 ... 00027) · 27 filings, 2022	PRIORITY

The narrow, defensible magnet novelty — the specific conductor and the digital-twin winding optimisation, not high-field REBCO as a category — is the commercial core that can earn ahead of any $Q > 1$ milestone, with markets in fusion magnet supply, MRI/NMR, accelerators, and proton therapy. The claim is scoped honestly: the high field is a *system field* result, not a stand-alone-coil record; the small-bore plug coil is structurally infeasible as a bare winding but resolved by a stress-managed structural shield (feasible-pending-FEA); and the winding-tape experiment returned a *null result*. The value is the method, not a field record.

THE TEAM

Founding partners, board, advisors & auditors

Kronos is built by a bench of advanced-fuel-fusion, high-field-magnet, direct-conversion, and materials specialists, with a board and operations team drawn from defense, national laboratories, and industry. Dates are shown as ranges; where a tenure has ended or is term-ending, the range says so.

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Legal counsel is retained; those roles are held on the internal roster and are not listed here.

LIST OF FIGURES

List of Figures

Figure 1 — The three tiers: physics-informed AI (deployable now, clamped), quantum sensing (near-term... 6

Figure 2 — The control-barrier safety clamp: 0 of 5000 trajectories escape the safe set with the filter, against... 7

Figure 3 — The quantum resource frontier: the linear kinetic operator is an early-fault-tolerant target; the... 8

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COLOPHON

How this document was made

The Kronos Fleet — Combined Editorial, 2026 is set in **Fraunces** (display), **Gelasio** (text), and **IBM Plex Mono** (data & equations), carried forward from the Kronos editorial design system.

The figures are rendered at 300 dpi from the deposited generator scripts; the document is composed as a single self-contained file with embedded fonts and imagery, paginated to US Letter, and rendered to PDF through a headless Chromium engine in matched light and dark editions.

Every headline quantity carries an evidence class; withdrawn and restated values are marked as such. Nothing herein is frozen beyond the founder-approved Tier-A set.

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Numbers and their classes. Quantities are reported with evidence classes and case labels; modelled, assumed, and requirement-class values are identified as such and are not to be quoted without their labels.

Public edition. This edition carries the physics and design only; all financial, funding, and defense-commercial content has been removed for public distribution. The scientific text corresponds to the arXiv and journal submissions.

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THE 2026 KRONOS SET

One programme, in five papers

This editorial is one of five that together describe the Kronos fusion programme — a breeder that funds a generator, and the magnet, conversion, and control science that enable both. Each is published open-access with a reproducible companion deposit.

PAPER I	Hyperion — The Tritium Foundry · 10.5281/zenodo.21746157
PAPER II	Aegis & MetroVolt — The Generator · 10.5281/zenodo.21746479
PAPER III	High-Field REBCO Magnets & Tape · 10.5281/zenodo.21842514
PAPER IV	Direct Energy Conversion · 10.5281/zenodo.21842864
PAPER V	AI, ML & Quantum Control (<i>this paper</i>) · 10.5281/zenodo.21842371

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